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Folding mechanism and electronic device

Abstract

This application discloses a folding mechanism and an electronic device. The folding mechanism includes a base, a first housing seat, a second housing seat, a first screen supporting plate, a second screen supporting plate, and a hinge mechanism. The first housing seat and the second housing seat are both rotatably connected to the base by means of the hinge mechanism, the first housing seat is rotatably arranged on the first screen supporting plate, and the second housing seat is rotatably arranged on the second screen supporting plate. First rotating portions and second rotating portions, distributed in an extension direction of the base, are respectively arranged at first ends of a first swing arm and a second swing arm.

Inventors: **Zhao; Dongqin (Dongguan, CN), Cheng; Dongcun (Dongguan, CN), Luo; Zhengjun (Dongguan, CN)**

Applicant: **VIVO MOBILE COMMUNICATION CO., LTD.** (Dongguan, CN)

Family ID: **1000008778291**

Assignee: **VIVO MOBILE COMMUNICATION CO., LTD. (Dongguan, CN)**

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Primary Examiner: Rathod; Abhishek M

Attorney, Agent or Firm: IPX PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2022/080909, filed Mar. 15, 2022, which claims priority to Chinese Patent Application No. 202110297137.8, filed Mar. 19, 2021. The entire contents of each of the above-referenced applications are expressly incorporated herein by reference.

TECHNICAL FIELD
(1) This application relates to the technical field of communication devices, and in particular to a folding mechanism and an electronic device.
BACKGROUND
(2) With the development of technology, the development of electronic devices becomes more and more rapid, and meanwhile, users' requirements for electronic devices become higher and higher. At present, flexible displays are also widely used in electronic devices, so that folding electronic devices are emerged.
(3) However, in the folding process, the bending areas of the flexible displays are easy to squeeze, and then excessive bending occurs, which will eventually affect the service life of the flexible displays.
SUMMARY
(4) Embodiments of this application aim to provide a folding mechanism and an electronic device.
(5) In a first aspect, an embodiment of this application discloses a folding mechanism, including a base, a first housing seat, a second housing seat, a first screen supporting plate and a second screen supporting plate, where the first housing seat is arranged on the first screen supporting plate, and the first screen supporting plate is in rotation fit with the first housing seat; the second housing seat is arranged on the second screen supporting plate, and the second screen supporting plate is in rotation fit with the second housing seat; the folding mechanism further includes a hinge mechanism, where the hinge mechanism includes a first swing arm, a second swing arm, a third swing arm and a fourth swing arm, the first swing arm and the third swing arm are both arranged on a same side as the first housing seat, and the second swing arm and the fourth swing arm are both arranged on a same side as the second housing seat; first rotating portions and second rotating portions, distributed in an extension direction of the base, are respectively arranged at first ends of the first swing arms and second swing arms, third rotating portions and fourth rotating portions are respectively arranged on two opposite sides of the base, the first rotating portions are in rotation fit with the third rotating portions, the second rotating portions are in rotation fit with the fourth rotating portions, and the first swing arm and the second swing arm are both limited to the base in a direction perpendicular to the extension direction of the base; a second end of the first swing arm is rotatably connected to the first housing seat, a first end of the third swing arm is rotatably connected to the base, a second end of the third swing arm is in sliding fit with the first housing seat, the second end of the third swing arm is configured to slide relative to and be in rotation fit with the first screen supporting plate, and a rotating axis of the first end of the first swing arm and a rotating axis of the first end of the third swing arm are distributed at an interval; and a second end of the second swing arm is rotatably connected to the second housing seat, a first end of the fourth swing arm is rotatably connected to the base, a second end of the fourth swing arm is in sliding fit with the second housing seat, the second end of the fourth swing arm is configured to slide relative to and be in rotation fit with the second screen supporting plate, and a rotating axis of the first end of the second swing arm and a rotating axis of the first end of the fourth swing arm are distributed at an interval.

(6) In a second aspect, an embodiment of this application discloses an electronic device, including the folding mechanism according to the first aspect, a flexible display, a first housing and a second housing, where the first housing is fixedly connected to the first housing seat, the second housing is fixedly connected to the second housing seat, the first screen supporting plate is arranged between the first housing and the base, the second screen supporting plate is arranged between the second housing and the base, and the flexible display is arranged on the base, the first housing, the second housing, the first screen supporting plate and the second screen supporting plate.
(7) The folding electronic device according to this embodiment of this application is folded inwards, that is, after the folding electronic device is folded, the flexible display is superimposed between the first folding portion and the second folding portion. In the folding process of the folding electronic device, the rotating axis of the first end of the first swing arm and the rotating axis of the first end of the third swing arm are distributed at an interval, and the rotating axis of the first end of the second swing arm and the rotating axis of the first end of the fourth swing arm are distributed at an interval. Therefore, when a user controls the first folding portion to rotate relative to the second folding portion, the first housing seat and the second housing seat oppositely rotate relative to the base, the second end of the first swing arm and the second end of the third swing arm relatively rotate, and the second end of the second swing arm relatively rotates relative to the second end of the fourth swing arm. Finally, the second end of the third swing arm drives the first screen supporting plate to rotate relative to the first housing seat, and the second end of the fourth swing arm drives the second screen supporting plate to rotate relative to the second housing seat. In this case, an end of the first screen supporting plate close to the base and an end of the second screen supporting plate close to the base are far away from each other, and finally, an accommodation space gradually expanded in a direction close to the base can be formed, so that an enough space is provided for the bending area of the flexible display in the folding state, and the bending area of the flexible display is prevented from being excessively squeezed and thus damaged.
(8) In this process, the second end of the third swing arm slides relative to the first housing seat, which can adapt to the rotation of the first housing seat driven by the second end of the first swing arm and prevent the rotation of the first housing seat from being locked. Similarly, the second end of the fourth swing arm slides relative to the second housing seat, which can adapt to the rotation of the second housing seat driven by the second end of the second swing arm and prevent the rotation of the second housing seat from being locked.
(9) Meanwhile, the first housing seat is rotatably connected to the first swing arm, and the second housing seat is rotatably connected to the second swing arm. Therefore, when the user controls the folding electronic device to fold, the first housing seat can rotate relative to the first swing arm, and the second housing seat can rotate relative to the second swing arm, so that the first housing connected to the first housing seat and the second housing connected to the second housing seat can further rotate to close and parallel positions, thereby preventing a tapered gap from being generated between the first folding portion and the second folding portion after the folding electronic device is folded, and thus preventing foreign matter from entering.
(10) In addition, the first rotating portions and the second rotating portions are respectively arranged between the first swing arm and the second swing arm, and the third rotating portions and the fourth rotating portions are correspondingly arranged on the two opposite sides of the base. The first rotating portions cooperate with the third rotating portions, and the second rotating portions cooperate with the fourth rotating portions, so that the first housing seat and the second housing seat are limited to the base in the direction perpendicular to the extension direction of the base while the first housing seat and the second housing seat are rotatably connected by means of the base, thereby reducing the connecting difficulty between the first housing seat and the second housing seat.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The accompanying drawings described herein are used to provide a further understanding of this application, and form part of this application. Exemplary embodiments of this application and descriptions thereof are used to explain this application, and do not constitute any inappropriate limitation to this application. In the accompanying drawings:

(2) FIG. **1** is a schematic structural diagram of a folding mechanism disclosed by an embodiment of this application;

(3) FIG. **2** is a schematic exploded diagram of the folding mechanism disclosed by an embodiment of this application;

(4) FIG. **3** is a schematic enlarged diagram of the folding mechanism including a base disclosed by an embodiment of this application;

(5) FIG. **4** is a schematic cross-sectional diagram of the folding mechanism in an unfolding state disclosed by an embodiment of this application;

(6) FIG. **5** is a schematic cross-sectional diagram of the folding mechanism in a certain state between the unfolding state and a folding state disclosed by an embodiment of this application; and

(7) FIG. **6** is a schematic cross-sectional diagram of the folding mechanism in the folding state disclosed by an embodiment of this application.

DETAILED DESCRIPTION

(8) The technical solutions in embodiments of this application are described in the following with reference to the accompanying drawings in the embodiments of this application. Apparently, the described embodiments are merely some rather than all of the embodiments of this application. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of this application without making creative efforts shall fall within the protection scope of this application.

(9) Terms “first” and “second” in the specification and claims of this application are used to distinguish similar objects, but are unnecessarily used to describe a specific sequence or order. It should be understood that data used in this way is exchangeable in a proper case, so that the embodiments of this application can be implemented in an order different from the order shown or described herein. Moreover, the objects distinguished by the terms “first” and “second” are usually of the same category, and the number of the objects is not limited, for example, the first object can be one or more. In addition, “and/or” used in the specification and claims represents at least one of the connected objects, and the character “/” generally indicates that the associated objects are in an “or” relationship.

(10) The folding mechanism and electronic device provided by the embodiments of this application will be described in detail below with reference to the drawings through specific embodiments and application scenarios thereof.

(11) As shown in FIG. **1** to FIG. **6**, an embodiment of this application discloses a folding mechanism. The disclosed folding mechanism can be applied to a folding electronic device, so that the folding and unfolding of the folding electronic device are achieved.

(12) The folding mechanism disclosed by this embodiment of this application includes a base **110**, a first housing seat **310**, a second housing seat **320**, a first screen supporting plate **210**, a second screen supporting plate **220** and a hinge mechanism.

(13) The base **110** is a fundamental component of the folding mechanism. The base **110** can provide a mounting base for other components of the folding mechanism, so that the other components of the folding mechanism are directly or indirectly mounted on the basis of the base **110**. In some embodiments, the base **110** is a sheet so as to reduce the occupied space. In some embodiments, the base **110** can be a metal sheet. The metal sheet can make the base **110** still have enough strength at

a small thickness, so that the support performance of the base **110** is not affected.
(14) The folding electronic device according to this embodiment of this application includes a first folding portion, a second folding portion and a flexible display.
(15) In the case that the folding mechanism is in a folding state, the folding electronic device is also in the folding state, so that the folding electronic device is in a portable state. At the moment, the first folding portion is superimposed with the second folding portion, and the flexible display is folded through its own deformation.
(16) In the case that the folding mechanism is in a unfolding state, the folding electronic device is also in the unfolding state, and at the moment, the first folding portion and the second folding portion are spread out, so that the flexible display is unfolded through its own deformation, thereby achieving the purpose of a large display area.
(17) The first folding portion is rotatably connected to the second folding portion by means of the folding mechanism, and the first folding portion rotates relative to the second folding portion, so that the folding electronic device is switched between the unfolding state and the folding state. That is to say, the first folding portion and the second folding portion are both connected to the folding mechanism.
(18) The first folding portion includes a first housing, and the second folding portion includes a second housing. The first housing seat **310** is configured to be fixedly connected to the first housing, so that the assembly connection between the folding mechanism and the first housing is achieved, and thus the assembly connection between the folding mechanism and the first folding portion is achieved. In some embodiments, the first housing seat **310** is connected to the first housing by using threaded connectors and fixed clamping, bonding, riveting and other fixing methods. This embodiment of this application does not limit the specific fixing method between the first housing seat **310** and the first housing.
(19) Similarly, the second housing seat **320** is configured to be fixedly connected to the second housing, so that the assembly connection between the folding mechanism and the second housing is achieved, and thus the assembly connection between the folding mechanism and the second folding portion is achieved. In some embodiments, the second housing seat **320** is connected to the second housing by using threaded connectors and fixed clamping, bonding, riveting and other fixing methods. This embodiment of this application does not limit the specific fixing method between the second housing seat **320** and the second housing.
(20) As described above, the folding electronic device includes a flexible display. The first screen supporting plate **210** and the second screen supporting plate **220** are structural members of the folding mechanism for supporting the flexible display. In some embodiments, the base **110** can also support the flexible display. The first screen supporting plate **210** and the second screen supporting plate **220** can cooperate with the first housing and the second housing to support the whole flexible display.
(21) The first housing seat **310** is arranged on the first screen supporting plate **210**, and the first screen supporting plate **210** is in rotation fit with the first housing seat **310**. In some embodiments, the first housing seat **310** is arranged on a back surface of the first screen supporting plate **210**. The back surface of the first screen supporting plate **210** is opposite to a supporting surface of the first screen supporting plate **210**. Therefore, the assembly position can avoid the effect of the first housing seat **310** on the supporting function of the first screen supporting plate **210**.
(22) The second housing seat **320** is arranged on the second screen supporting plate **220**, and the second screen supporting plate **220** is in rotation fit with the second housing seat **320**. In some embodiments, the second housing seat **320** is arranged on a back surface of the second screen supporting plate **220**. The back surface of the second screen supporting plate **220** is opposite to a supporting surface of the second screen supporting plate **220**. Therefore, the assembly position can avoid the effect of the second housing seat **320** on the supporting function of the second screen supporting plate **220**.

(23) In this embodiment of this application, the first housing seat **310** and the first screen supporting plate **210** are arranged on a first side of the base **110** so as to correspond to the first folding portion. The second housing seat **320** and the second screen supporting plate **220** are arranged on a second side of the base **110** so as to correspond to the second folding portion. It should be noted that: the first side and the second side are oppositely arranged.
(24) In some embodiments, the first housing seat **310** and the second housing seat **320** can be symmetrically arranged on two sides of the base **110**. Similarly, the first screen supporting plate **210** and the second screen supporting plate **220** can also be symmetrically arranged on the two sides of the base **110**. The symmetrical arrangement enables the weight of the folding mechanism to be balanced, and avoids a bias phenomenon.
(25) The hinge mechanism disclosed by this embodiment of this application includes a first swing arm **510**, a second swing arm **520**, a third swing arm **530** and a fourth swing arm **540**.
(26) The first swing arm **510** and the third swing arm **530** are both arranged on a same side as the first housing seat **310**, i.e., the first swing arm **510**, the third swing arm **530** and the first housing seat **310** are all located on the first side of the base **110**.
(27) The second swing arm **520** and the fourth swing arm **540** are both arranged on a same side as the second housing seat **320**, i.e., the second swing arm **520**, the fourth swing arm **540** and the second housing seat **320** are all located on the second side of the base **110**.
(28) First rotating portions **501** and second rotating portions **502** are respectively arranged at first ends of the first swing arm **510** and the second swing arm **520**, the first rotating portions **501** and the second rotating portions **502** are distributed in an extension direction of the base **110**, and correspondingly, third rotating portions **111** and fourth rotating portions **112** are respectively arranged on two opposite sides of the base **110**. In some embodiments, the extension direction of the base **110** can be a direction A in FIG. **1**, the two opposite sides of the base **110** are in the other direction perpendicular to the direction A, and the two opposite sides of the base **110** can be the two sides of the base **110** opposite in a direction B in FIG. **4**.
(29) The first rotating portions **501** are in rotation fit with the third rotating portions **111**, the second rotating portions **502** are in rotation fit with the fourth rotating portions **112**, and the first swing arm **510** and the second swing arm **520** are limited to the base **110** in a direction perpendicular to the extension direction of the base **110** by means of the corresponding fitting first rotating portions **501** and third rotating portions **111** and the corresponding fitting second rotating portions **502** and fourth rotating portions **112**. That is to say, after the first swing arm **510** and the second swing arm **520** are in rotation fit with the base **110**, the first swing arm **510** and the second swing arm **520** can be connected into a whole by means of the base **110**.
(30) The second end of the first swing arm **510** is rotatably connected to the first housing seat **310**. In some embodiments, the second end of the first swing arm **510** can be hinged to the first housing seat **310** by means of the third rotating shaft **353** so as to achieve a rotational connection.
(31) The first end of the third swing arm **530** is rotatably connected to the base **110**, so that the third swing arm **530** can rotate around the base **110** by means of the first end of the third swing arm. The second end of the third swing arm **530** is in sliding fit with the first housing seat **310**, so that the second end of the third swing arm **530** is configured to slide relative to the first housing seat **310**. The second end of the third swing arm **530** is configured to slide relative to and be in rotation fit with the first screen supporting plate **210**, so that the second end of the third swing arm **530** swings to drive the first screen supporting plate **210** to rotate relative to the first housing seat **310** while the assembly connection between the third swing arm **530** and the first screen supporting plate **210** is achieved.
(32) A rotating axis of the first end of the first swing arm **510** and a rotating axis of the first end of the third swing arm **530** are distributed at an interval, so that in a swinging process of the first swing arm **510** and the third swing arm **530**, the second end of the first swing arm **510** and the second end of the third swing arm **530** can relatively rotate, thereby driving the first screen

supporting plate **210** to rotate relative to the first housing seat **310**. In some embodiments, the rotating axis of the first end of the first swing arm **510** is parallel to the rotating axis of the first end of the third swing arm **530**, so that the stability of the first swing arm **510** and the third swing arm **530** during rotation can be improved well.
(33) The second end of the second swing arm **520** is rotatably connected to the second housing seat **320**. In some embodiments, the second end of the second swing arm **520** can be hinged to the second housing seat **320** by means of the fourth rotating shaft **354** so as to achieve a rotational connection.
(34) The first end of the fourth swing arm **540** is rotatably connected to the base **110**, so that the fourth swing arm **540** can rotate around the base **110** by means of the first end of the fourth swing arm. The second end of the fourth swing arm **540** is in sliding fit with the second housing seat **320**, so that the second end of the fourth swing arm **540** is configured to slide relative to the second housing seat **320** during rotation. The second end of the fourth swing arm **540** is configured to slide relative to and be in rotation fit with the second screen supporting plate **220**, so that the second end of the fourth swing arm **540** swings to drive the second screen supporting plate **220** to rotate relative to the second housing seat **320** while the assembly connection between the fourth swing arm **540** and the second screen supporting plate **220** is achieved.
(35) The rotating axis of the first end of the second swing arm **520** and the rotating axis of the first end of the fourth swing arm **540** are distributed at an interval, so that in the swinging process of the second swing arm **520** and the fourth swing arm **540**, the second end of the second swing arm **520** and the second end of the fourth swing arm **540** can relatively rotate, thereby driving the second screen supporting plate **220** to rotate relative to the second housing seat **320**. In some embodiments, the rotating axis of the first end of the second swing arm **520** is parallel to the rotating axis of the first end of the fourth swing arm **540**, so that the stability of the second swing arm **520** and the fourth swing arm **540** during rotation can be improved well.
(36) The folding electronic device according to this embodiment of this application is folded inwards, that is, after the folding electronic device is folded, the flexible display is superimposed between the first folding portion and the second folding portion. In the folding process of the folding electronic device, the rotating axis of the first end of the first swing arm **510** and the rotating axis of the first end of the third swing arm **530** are distributed at an interval, and the rotating axis of the first end of the second swing arm **520** and the rotating axis of the first end of the fourth swing arm **540** are distributed at an interval. Therefore, when a user controls the first folding portion to rotate relative to the second folding portion, the first housing seat **310** and the second housing seat **320** oppositely rotate relative to the base **110**, the second end of the first swing arm **510** and the second end of the third swing arm **530** relatively rotate, the second end of the second swing arm **520** relatively rotates relative to the second end of the fourth swing arm **540**, finally, the second end of the third swing arm **530** drives the first screen supporting plate **210** to rotate relative to the first housing seat **310**, and the second end of the fourth swing arm **540** drives the second screen supporting plate **220** to rotate relative to the second housing seat **320**. In this case, an end of the first screen supporting plate **210** close to the base **110** and an end of the second screen supporting plate **220** close to the base **110** are far away from each other, and finally, an accommodation space gradually expanded in the direction close to the base **110** can be formed, so that an enough space is provided for the bending area of the flexible display in the folding state, and the bending area of the flexible display is prevented from being excessively squeezed and thus damaged.
(37) In this process, the second end of the third swing arm **530** slides relative to the first housing seat **310**, which can adapt to the rotation of the first housing seat **310** driven by the second end of the first swing arm **510** and prevent the rotation of the first housing seat **310** from being locked. Similarly, the second end of the fourth swing arm **540** slides relative to the second housing seat **320**, which can adapt to the rotation of the second housing seat **320** driven by the second end of the

second swing arm **520** and prevent the rotation of the second housing seat **320** from being locked.
(38) Meanwhile, the first housing seat **310** is rotatably connected to the first swing arm **510**, and the second housing seat **320** is rotatably connected to the second swing arm **520**. Therefore, when the user controls the folding electronic device to fold, the first housing seat **310** can rotate relative to the first swing arm **510**, and the second housing seat **320** can rotate relative to the second swing arm **520**, so that the first housing connected to the first housing seat **310** and the second housing connected to the second housing seat **320** can further rotate to close and parallel positions, thereby preventing a tapered gap from being generated between the first folding portion and the second folding portion after the folding electronic device is folded, and thus preventing foreign matter from entering.
(39) In addition, the first rotating portions **501** and the second rotating portions **502** are respectively arranged between the first swing arm **510** and the second swing arm **520**, and the third rotating portions **111** and the fourth rotating portions **112** are correspondingly arranged on the two opposite sides of the base **110**. The first rotating portions **501** cooperate with the third rotating portions **111**, and the second rotating portions **502** cooperate with the fourth rotating portions **112**, so that the first housing seat **310** and the second housing seat **320** are limited to the base **110** in the direction perpendicular to the extension direction of the base **110** while the first housing seat **310** and the second housing seat **320** are rotatably connected by means of the base **110**, thereby reducing the connecting difficulty between the first housing seat **310** and the second housing seat **320**.
(40) In some embodiments, the first rotating portions **501** and the third rotating portions **111** as well as the second rotating portions **502** and the fourth rotating portions **112** can be all corresponding shaft hole type structural members. In some embodiments, the first rotating portions **501**, the second rotating portions **502**, the third rotating portions **111** and the fourth rotating portions **112** are all provided with corresponding curved fitting members. That is to say, the first rotating portions **501** and the third rotating portions **111** are provided with corresponding curved fitting surfaces, so that the first rotating portions are in rotation fit with the third rotating portions. Correspondingly, corresponding curved fitting surfaces are arranged between the second rotating portions **502** and the fourth rotating portions **112**, so that the second rotating portions **502** are in rotation fit with the fourth rotating portions **112**. In some embodiments, the curved fitting surfaces on the first rotating portions **501** and the second rotating portions **502** may have the same structure or different structures, which is not limited herein.
(41) Meanwhile, in order to ensure that the first housing seat **310** and the base **110** which are rotatably connected are limited to each other in the direction perpendicular to the extension direction of the base **110** by means of the first rotating portions **501** and the third rotating portions **111** as well as the second rotating portions **502** and the fourth rotating portions **112**, correspondingly, the curved fitting surfaces of the first rotating portions **501** and the curved fitting surfaces of the second rotating portions **502** are oppositely arranged, if the arc of the curved fitting surfaces of the first rotating portions **501** is arranged upwards, the arc of the curved fitting surfaces of the second rotating portions **502** can be arranged downwards, or one arc is arranged leftwards, and the other arc is arranged rightwards. In the case of the described technical solution, the third rotating portions **111** cooperate with the first rotating portions **501**, and the fourth rotating portions **112** cooperate with the second rotating portions **502**, so that the first housing seat **310** and the base **110** are limited to each other in the direction perpendicular to the extension direction of the base **110**.
(42) Correspondingly, in the case that the first rotating portions **501** and the second rotating portions **502** on the first swing arm **510** use the described technical solution, the first rotating portions **501** and the second rotating portions **502** on the second swing arm **520** can also be arranged according to the described technical solution. In some embodiments, the size and position relationship of the first rotating portions **501** and the second rotating portions **502** arranged on the second swing arm **520** can be correspondingly the same as or different from the size and position

relationship of the first rotating portions **501** and the second rotating portions **502** arranged on the first swing arm **510**, which is not limited herein.
(43) In order to improve the stability of the cooperative relationship between the first housing seat **310** and the second housing seat **320** and the base **110**, as shown in FIG. **3**, at least one of the first swing arm **510** and the second swing arm **520** is provided with two first rotating portions **501**, and at least one second rotating portion **502** is arranged between the two first rotating portions **501**. The first rotating portions **501** and the second rotating portions **502** apply opposite limiting directions to the first swing arm **510**. Therefore, by using the described technical solution, the first rotating portions **501** are respectively arranged on two opposite sides of the second rotating portions **502**, so that the case of mutual tilting between the first swing arm **510** and the base **110** can be prevented, and the fitting stability between the first housing seat **310** and the base **110** is improved. In order to ensure that the fitting stability between all the components of the whole folding mechanism is relatively good, both the first swing arm **510** and the second swing arm **520** can use the described technical solution.
(44) Furthermore, at least one of the first swing arm **510** and the second swing arm **520** is provided with a plurality of first rotating portions **501** and a plurality of second rotating portions **502**, and the first rotating portions **501** and the second rotating portions **502** are alternately distributed. In the case of the described technical solution, the case of mutual tilting in the relative rotating process of the first swing arm **510** and the base **110** can be further prevented, so that the fitting stability between the first swing arm **510** and the base **110** is further improved. In some embodiments, both the first swing arm **510** and the second swing arm **520** can use the described technical solution, so that the cooperative relationship between the first swing arm **510** and the second swing arm **520** and the base **110** is stable, and thus the first screen supporting plate **210** and the second screen supporting plate **220** in the folding mechanism achieve reliable and good supporting effects on the flexible display.
(45) In some embodiments, the first swing arm **510** and the second swing arm **520** are both respectively provided with two first rotating portions **501** and two second rotating portions **502**, and the two first rotating portions **501** and the two second rotating portions **502** are alternately distributed. In addition, two third rotating portions **111** and fourth rotating portions **112** are respectively arranged on the two opposite sides of the base **110** corresponding to one first swing arm **510** and one second swing arm **520**. In the case of the described technical solution, the first swing arm **510** and the second swing arm **520** are basically in stable rotation fit with the base **110**, and the total number of structures on all the components is relatively small, so that the processing difficulty and production cost can be reduced to a certain extent.
(46) As described above, the first rotating portions **501** and the second rotating portions **502** are both respectively provided with the curved fitting surfaces, by taking the first swing arm **510** as an example, in the first swing arm **510**, the first rotating portions **501** are provided with matching grooves, the second rotating portions are provided with matching protrusions, and the matching grooves and the matching protrusions are both provided with curved fitting surfaces. In some embodiments, by using etching or drilling methods, the matching grooves and the matching protrusions provided with the curved fitting surfaces can be formed on the base **110**, correspondingly, one of the third rotating portions **111** and the fourth rotating portions **112** on the base **110** can be provided with matching grooves, the other can be provided with matching protrusions, the matching grooves of the base **110** are in corresponding fit with the matching protrusions on the first swing arm **510**, and the matching protrusions on the bas portion **110** are in corresponding fit with the matching grooves on the first swing arm **510**, so that the first swing arm and the base are limited to each other in the direction perpendicular to the extension direction of the base **110** while the first swing arm **510** is in rotation fit with the base **110**.
(47) In the case of the described structure, in order to ensure that the first swing arm **510** can be in normal fit with the base **110**, all the matching grooves need to have notches, so that the

corresponding structures in the first swing arm **510** extend into the matching grooves from the notches of the matching grooves on the base **110**; meanwhile, the corresponding structures on the base **110** can extend into the matching grooves through the notches of the matching grooves on the first swing arm **510**, so that the purpose of rotatably connecting the first swing arm **510** to the base **110** is achieved.
(48) In some embodiments, the matching protrusions can be arranged on the side towards which the notches of the matching grooves face. In another embodiment of this application, as shown in FIG. **3**, the matching protrusions can be arranged away from the notches of the matching grooves. In this case, according to FIG. **3**, the matching protrusions on the base **110** can be arranged in the direction towards which the matching grooves face, so that when the third rotating portions **111** and the fourth rotating portions **112** of the same size are formed, by using the described technical solution, the occupied space of the third rotating portions **111** and the fourth rotating portions **112** on the base **110** can be greatly reduced, and the difficulty and workload of forming the third rotating portions **111** and the fourth rotating portions **112** can be reduced to a certain extent.
(49) In some embodiments, as shown in FIG. **3**, the first swing arm **510** is opposite to the second swing arm **520**, and the third swing arm **530** is opposite to the fourth swing arm **540**. In some embodiments, the structures of the first swing arm **510** and the second swing arm **520** can be symmetrical with each other, and correspondingly, the third rotating portions **111** and the fourth rotating portions **112** which are arranged on the two opposite sides of the base **110** and respectively cooperate with the first swing arm **510** and the second swing arm **520** are correspondingly arranged. Similarly, the structures of the third swing arm **530** and the fourth swing arm **540** can be symmetrical with each other, and the structures on the base **110** which cooperate with the third swing arm and the fourth swing arm are also symmetrical with each other. By using the described technical solution, the symmetry of the folding mechanism can be greatly improved, thus the folding and unfolding performance of the folding structure is improved, the synchrony of the first screen supporting plate **210** and the second screen supporting plate **220** is good, and the supporting effect on the flexible display is improved.
(50) As described above, the second end of the third swing arm **530** is configured to slide relative to the first housing seat **310**. On this basis, in order to improve the matching effect, in some embodiments, the first housing seat **310** can be provided with a first sliding groove **311**, and the second end of the third swing arm **530** is in sliding fit with the first sliding groove **311**.
(51) In a further technical solution, an inner wall of the first sliding groove **311** is provided with a first avoidance hole, a first track body **230** is fixed to the first screen supporting plate **210**, the second end of the third swing arm **530** is provided with a first sliding member **551**, the first track body **230** passes through the first avoidance hole and at least partially extends into the first sliding groove **311**, and the first sliding member **551** is configured to slide relative to and be in rotation fit with the first track body **230**. The structure is relatively simple, and can meet the assembly requirement that the second end of the third swing arm **530** is configured to slide relative to and be in rotation fit with the first screen supporting plate **210**. In some embodiments, the first track body **230** is of a curve structure, and in the rotating process of the third swing arm **530**, the first sliding member **551** can slide in the first track body **230**. According to the structure, the structure of the first sliding groove **311** can be fully used, and a bottom wall of the first sliding groove **311** is provided with the first avoidance hole for assembly, so that the assembly structure is more compact.
(52) Similarly, the second housing seat **320** is provided with a second sliding groove **321**, and the second end of the fourth swing arm **540** is in sliding fit with the second sliding groove **321**. By using the structure, the sliding fit stability can be improved.
(53) In a further technical solution, an inner wall of the second sliding groove **321** is provided with a second avoidance hole, a second track body is fixed to the second screen supporting plate **220**, the second end of the fourth swing arm **540** is provided with a second sliding member **552**, the

second track body passes through the second avoidance hole and at least partially extends into the second sliding groove **321**, and the second sliding member **552** is configured to slide relative to and be in rotation fit with the second track body.
(54) As described above, the first screen supporting plate **210** can be in rotation fit with the first housing seat **310** by means of structural components such as a shaft hole mechanism, the first screen supporting plate **210** is provided with a first shaft seat **250**, the first housing seat **310** is provided with a second shaft seat **330**, and the first shaft seat **250** and the second shaft seat **330** are connected by means of a first rotating shaft **351**. The described rotational connecting mechanism has the advantages of a simple structure, convenience in assembly and relatively high reliability. In some embodiments, a number of the first shaft seat **250** and the second shaft seat **330** can be both one, and in this case, the position of the first rotating shaft **351** can be limited by means of a structural member such as a shaft pin; or, a number of at least one of the first shaft seat **250** and the second shaft seat **330** can be more, and in this case, the reliability of a rotational connection between the first housing seat **310** and the first screen supporting plate **210** can be improved. In some embodiments, a plurality of groups of first shaft seats **250** and second shaft seats **330** can be arranged, any group includes at least one first shaft seat **250** and at least one second shaft seat **330**, and in this case, the reliability of a rotation fit between the first screen supporting plate **210** and the first housing seat **310** can be further improved.
(55) Furthermore, as shown in FIG. **2**, the first screen supporting plate **210** can be provided with a first avoidance gap **211**, at least a part of the second shaft seat **330** is arranged in the first avoidance gap **211**, and in this case, the occupied space of the first screen supporting plate **210** and the first housing seat **310** in the thickness direction of the first screen supporting plate **210** can be reduced, so that the thickness of the whole folding mechanism is reduced, and thus the thickness of the electronic device using the folding mechanism is also reduced relatively.
(56) On the basis of the described embodiment, the second screen supporting plate **220** is provided with a third shaft seat, the second housing seat **320** is provided with a fourth shaft seat **340**, and the third shaft seat and the fourth shaft seat **340** are connected by means of a second rotating shaft **352**. The described rotational connecting mechanism has the advantages of a simple structure, convenience in assembly and relatively high reliability. In some embodiments, a number of the third shaft seat and the fourth shaft seat **340** can be both one, and in this case, the position of the second rotating shaft **352** can be limited by means of a structural member such as a shaft pin; or, a number of at least one of the third shaft seat and the fourth shaft seat **340** can be more, and in this case, the reliability of a rotational connection between the second housing seat **320** and the second screen supporting plate **220** can be improved. In some embodiments, a plurality of groups of third shaft seats and fourth shaft seats **340** can be arranged, any group includes at least one third shaft seat and at least one fourth shaft seat **340**, and in this case, the reliability of a rotation fit between the second screen supporting plate **220** and the second housing seat **320** can be further improved.
(57) Furthermore, as shown in FIG. **2**, the second screen supporting plate **220** can be provided with a second avoidance gap **221**, at least a part of the fourth shaft seat **340** is arranged in the second avoidance gap **221**, and in this case, the occupied space of the second screen supporting plate **220** and the second housing seat **320** in the thickness direction of the second screen supporting plate **220** can be reduced, so that the thickness of the whole folding mechanism is reduced, and thus the thickness of the electronic device using the folding mechanism is also reduced relatively.
(58) In the folding mechanism disclosed by this embodiment of this application, a number of the hinge mechanisms can be at least two, and the at least two hinge mechanisms are arranged at an interval. According to the structure, stable rotational connection assembly can be achieved by means of a plurality of hinge mechanisms.
(59) In some embodiments, as shown in FIG. **3** and FIG. **6**, the folding mechanism further includes a cover body **120**, where the cover body **120** is fixedly connected to a side of the base **110** away from the first screen supporting plate **210**, the cover body **120** achieves a protective effect on the

base **110**, and the cover body **120** can shield the folding mechanism, so that when the folding mechanism is applied to the electronic device, a number of exposed parts of the electronic device is reduced as far as possible, on one hand, the appearance performance of the electronic device is improved, on the other hand, the dustproof and waterproof performance of the electronic device is improved, and the reliability of the electronic device is improved. The cover body **120** can be fixed to the base **110** by using a bonding method. In the other embodiment of this application, the base **110** and the cover body **120** can be detachably and fixedly connected by means of threaded connectors such as screws, so that the reliability in connection between the base and the cover body is improved.

(60) Meanwhile, in the folding state, a projection of the cover body **120** in a plane where the distribution direction of the first screen supporting plate **210** and the second screen supporting plate **220** and the extension direction of the base **110** are located covers the first housing seat **310**, the second housing seat **320** and the base **110**, so that the cover body **120** achieves a good shielding effect. It should be noted that: the distribution direction of the first screen supporting plate **210** and the second screen supporting plate **220** can be the direction B in FIG. **4**.

(61) In some embodiments, according to the specific structure and size of other components in the folding mechanism, the structure, size and mounting position of the cover body **120** are designed, so that when the folding mechanism is viewed from the front of the side where the cover body **120** is located in the folding state of the folding mechanism, the first housing seat **310**, the second housing seat **320** and the base **110** in the folding mechanism can be shielded by the cover body **120**.

(62) Furthermore, as shown in FIG. **4** to FIG. **6**, the first swing arm **510** and/or the second swing arm **520** includes an avoidance structure, and the avoidance structure is provided with an avoidance groove; and as shown in FIG. **4**, a side of the cover body **120** is accommodated into the avoidance groove in the unfolding state. In some embodiments, the position and size of the avoidance groove can be flexibly determined according to the extension position and thickness of the cover body **120**, which is not limited herein.

(63) As described above, the folding mechanism disclosed by this embodiment of this application has the folding state and the unfolding state; and in the folding state, a distance between the first end of the first screen supporting plate **210** and the first end of the second screen supporting plate **220** is a first distance, a distance between the second end of the first screen supporting plate **210** and the second end of the second screen supporting plate **220** is a second distance, the first distance is greater than the second distance, the second distance is smaller and is close to zero. In some embodiments, the second distance can be zero. The first end of the first screen supporting plate **210** is an end of the first screen supporting plate **210** close to the base **110**, the second end of the first screen supporting plate **210** is an other end of the second screen supporting plate **220** away from the base **110**, the first end of the second screen supporting plate **220** is an end of the second screen supporting plate **220** close to the base **110**, and the second end of the second screen supporting plate **220** is an other end of the screen supporting plate **220** away from the base **110**.

(64) In the unfolding state, a supporting surface of the first screen supporting plate **210** is coplanar with a supporting surface of the second screen supporting plate **220**, the supporting surface of the first screen supporting plate **210** and the back surface of the first screen supporting plate **210** are oppositely arranged, and the supporting surface of the second screen supporting plate **220** and the back surface of the second screen supporting plate **220** are oppositely arranged. In this case, in the folding state of the folding mechanism, the first screen supporting plate **210** and the second screen supporting plate **220** form the accommodation space gradually expanded in the direction close to the base **110**. In the unfolding state, a large supporting structure can be formed in a coplanar form, so that auxiliary supporting for the flexible display is achieved.

(65) On the basis of the folding mechanism disclosed by an embodiment of this application, an embodiment of this application discloses an electronic device, where the electronic device includes the folding mechanism described in the foregoing embodiment.

(66) The electronic device disclosed by this embodiment of this application further includes a first folding portion and a second folding portion, where the first folding portion includes a first housing, the second folding portion includes a second housing, the first housing is fixedly connected to the first housing seat **310**, the second housing is fixedly connected to the second housing seat **320**, the first screen supporting plate **210** is arranged between the first housing and the base **110**, the second screen supporting plate **220** is arranged between the second housing and the base **110**, and a flexible display is arranged on the base **110**, the first housing, the second housing, the first screen supporting plate **210** and the second screen supporting plate **220**.
(67) In some solutions, the first housing can be provided with a first groove, the second housing can be provided with a second groove, the first screen supporting plate **210** is located in the first groove, and the second screen supporting plate **220** is located in the second groove. When the electronic device is in the folding state, the first housing is superimposed with the second housing. The first screen supporting plate **210** and the second screen supporting plate **220** are respectively located in the first groove and the second groove. Therefore, a gap between the first housing and the second housing can be reduced after the electronic device is folded, and the first housing and the second housing almost attach to each other.
(68) The electronic device disclosed by this embodiment of this application can be a mobile phone, a computer, an ebook reader and a wearable device. This embodiment of this application does not limit the specific type of the electronic device.
(69) It should be noted that: the terms “include”, “comprise”, or any other variation thereof in this specification is intended to cover a non-exclusive inclusion, which specifies the presence of stated processes, methods, objects, or apparatuses, but do not preclude the presence or addition of one or more other processes, methods, objects, or apparatuses. Without more limitations, elements defined by the sentence “including one . . . ” does not exclude that there are still other same elements in the processes, methods, objects, or apparatuses. Furthermore, it should be pointed out that the scope of the methods and apparatuses in the embodiments of the present application is not limited to execute functions in the shown or discussed order, but may execute functions in a substantially simultaneous manner or in the reverse order according to the involved functions, for example, the described methods may be executed in a different order than described, and steps may be added, omitted, or combined. In addition, features described in some examples may also be combined in other examples.
(70) The embodiments of the present application have been described above with reference to the accompanying drawings. This application is not limited to the specific embodiments described above, and the specific embodiments described above are merely exemplary and not limitative. Those of ordinary skill in the art may make various variations under the teaching of this application without departing from the spirit of this application and the protection scope of the claims, and such variations shall all fall within the protection scope of this application.

Claims

1. A folding mechanism, comprising a base, a first housing seat, a second housing seat, a first screen supporting plate, and a second screen supporting plate, wherein: the first housing seat is arranged on the first screen supporting plate, and the first screen supporting plate is in rotation fit with the first housing seat; the second housing seat is arranged on the second screen supporting plate, and the second screen supporting plate is in rotation fit with the second housing seat; the folding mechanism further comprises a hinge mechanism, wherein the hinge mechanism comprises a first swing arm, a second swing arm, a third swing arm, and a fourth swing arm, the first swing arm and the third swing arm are both arranged on a same side as the first housing seat, and the second swing arm and the fourth swing arm are both arranged on a same side as the second housing seat; first rotating portions and second rotating portions, distributed in an extension direction of the

base, are respectively arranged at first ends of the first swing arms and second swing arms, third rotating portions and fourth rotating portions are respectively arranged on two opposite sides of the base, the first rotating portions are in rotation fit with the third rotating portions, the second rotating portions are in rotation fit with the fourth rotating portions, and the first swing arm and the second swing arm are both limited to the base in a direction perpendicular to the extension direction of the base; a second end of the first swing arm is rotatably connected to the first housing seat, a first end of the third swing arm is rotatably connected to the base, the first housing seat is provided with a first sliding groove, a second end of the third swing arm is in sliding fit with the first sliding groove, an inner wall of the first sliding groove is provided with a first avoidance hole, a first track body is fixed to the first screen supporting plate, the second end of the third swing arm is provided with a first sliding member, the first track body passes through the first avoidance hole and at least partially extends into the first sliding groove, the first sliding member is configured to slide relative to and be in rotation fit with the first track body, and a rotating axis of the first end of the first swing arm and a rotating axis of the first end of the third swing arm are distributed at an interval; and a second end of the second swing arm is rotatably connected to the second housing seat, a first end of the fourth swing arm is rotatably connected to the base, the second housing seat is provided with a second sliding groove, a second end of the fourth swing arm is in sliding fit with the second sliding groove, an inner wall of the second sliding groove is provided with a second avoidance hole, as second track body is fixed to the second screen supporting plate, the second end of the fourth swing arm is provided with a second sliding member, the second track body passes through the second avoidance hole and at least partially extends into the second sliding groove, the second sliding ember is configured to slide relative to and be in rotation fit with the second track body, and a rotating axis of the first end of the second swing arm and a rotating axis of the first end of the fourth swing arm are distributed at an interval.

2. The folding mechanism according to claim 1, wherein the first rotating portions and the second rotating portions as well as the third rotating portions and the fourth rotating portions are respectively provided with corresponding curved fitting surfaces, and the curved fitting surfaces of the first rotating portions and the curved fitting surfaces of the second rotating portions are oppositely arranged.

3. The folding mechanism according to claim 2, wherein at least one of the first swing arm and the second swing arm is provided with two first rotating portions, and at least one of the second rotating portions is arranged between the two first rotating portions.

4. The folding mechanism according to claim 3, wherein at least one of the first swing arm and the second swing arm is provided with a plurality of the first rotating portions and a plurality of the second rotating portions, and the first rotating portions and the second rotating portions are alternately distributed.

5. The folding mechanism according to claim 2, wherein the first rotating portions are provided with matching grooves, the second rotating portions comprise matching protrusions, the matching grooves and the matching protrusions are both provided with the curved fitting surfaces, and in the extension direction of the base, the matching protrusions deviate from notches of the matching grooves.

6. The folding mechanism according to claim 1, wherein the first swing arm is opposite to the second swing arm, and the third swing arm is opposite to the fourth swing arm.

7. The folding mechanism according to claim 1, wherein the first screen supporting plate is provided with a first shaft seat, the first housing seat is provided with a second shaft seat, and the first shaft seat and the second shaft seat are connected by means of a first rotating shaft.

8. The folding mechanism according to claim 7, wherein the first screen supporting plate is provided with a first avoidance gap, and at least a part of the second shaft seat is arranged in the first avoidance gap.

9. The folding mechanism according to claim 7, wherein the second screen supporting plate is

provided with a third shaft seat, the second housing seat is provided with a fourth shaft seat, and the third shaft seat and the fourth shaft seat are connected by means of a second rotating shaft.
10. The folding mechanism according to claim 9, wherein the second screen supporting plate is provided with a second avoidance gap, and at least a part of the fourth shaft seat is arranged in the second avoidance gap.
11. The folding mechanism according to claim 1, wherein a number of the hinge mechanisms is at least two, and the at least two hinge mechanisms are arranged at an interval.
12. The folding mechanism according to claim 1, wherein the folding mechanism further comprises a cover body, the cover body is fixedly connected to a side of the base away from the first screen supporting plate, and when the folding mechanism is in a folding state, a projection of the cover body in a plane where a rotating direction of the second rotating portions is located covers the first housing seat, the second housing seat, and the base.
13. The folding mechanism according to claim 12, wherein the first swing arm or the second swing arm comprises an avoidance structure, the avoidance structure is provided with an avoidance groove, and when the folding mechanism is in an unfolding state, a side of the cover body is accommodated into the avoidance groove.
14. The folding mechanism according to claim 1, wherein the folding mechanism has a folding state and an unfolding state, wherein: in the folding state, a distance between a first end of the first screen supporting plate and a first end of the second screen supporting plate is a first distance, a distance between a second end of the first screen supporting plate and a second end of the second screen supporting plate is a second distance, and the first distance is greater than the second distance, wherein the first end of the first screen supporting plate is an end of the first screen supporting plate close to the base, the second end of the first screen supporting plate is an other end of the first screen supporting plate away from the base, the first end of the second screen supporting plate is an end of the second screen supporting plate close to the base, and the second end of the second screen supporting plate is an other end of the screen supporting plate away from the base; and in the unfolding state, a supporting surface of the first screen supporting plate is coplanar with a supporting surface of the second screen supporting plate.
15. An electronic device, comprising the folding mechanism according to claim 1, a flexible display, a first housing, and a second housing, wherein: the first housing is fixedly connected to the first housing seat; the second housing is fixedly connected to the second housing seat; the first screen supporting plate is arranged between the first housing and the base; the second screen supporting plate is arranged between the second housing and the base; and the flexible display is arranged on the base, the first housing, the second housing, the first screen supporting plate, and the second screen supporting plate.
16. The electronic device according to claim 15, wherein the first housing is provided with a first groove, the second housing is provided with a second groove, the first screen supporting plate is arranged in the first groove, and the second screen supporting plate is arranged in the second groove.